

Alireza Laalmanesh

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Education

University of Waterloo

Master of Mathematics in Computational Mathematics (Co-op)

Waterloo, Canada

Sep. 2025 – Present

- **i-Comp-Data Fellowship:** Recipient of the Provost-funded inter-faculty award for Computation & Data research.
- Advisors: **Dr. David Del Rey Fernández** (Pratt & Whitney Chair in Industrial AI) & **Dr. Leia Minaker** (Director, Future Cities Institute).
- Focus: Numerical Analysis, Deep Learning, Urban Data Science, Optimization.

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Industrial Engineering & Information Systems

Tehran, Iran

2019 – 2024

- **GPA: 3.5/4.0** (Top 5% of class).
- Thesis: "Operating Room Scheduling using Meta-heuristic Optimization".

Technical Skills

Languages: Python, C/C++

AI & Data Science: PyTorch, scikit-learn, EfficientNet, CNNs, DeepInsight (Tabular→Image), t-SNE, XGBoost/GBM

Systems & Tools: Docker, Linux (Root/Kernel utilities), ADB, USB/IP, Git, Asyncio, Selenium, LaTeX

Math Concepts: Linear Algebra, Numerical Optimization, Loss Function Design, Stochastic Modeling, Calculus

Professional Experience

University of Waterloo

Graduate Teaching Assistant (Math Faculty)

Waterloo, Canada

Sep. 2025 – Present

- **Courses:** Applied Linear Algebra (MATH 106/136), Calculus (MATH 128), Algebra for Honours (MATH 135).
- Facilitated understanding of complex mathematical concepts including matrix operations, eigenvalues, and differential equations for 100+ students per term.
- Conducted weekly tutorials and office hours to simplify rigorous proof-based logic and computational applications.
- Evaluated examinations and assignments using **Crowdmark**, providing detailed technical feedback to students.

TAV Institute

Factory Layout Design Intern

Tehran, Iran

2024

- Optimized factory layouts to improve workflow efficiency using spatial analysis and process flow modeling.
- Conducted capacity analysis to identify bottlenecks in material handling systems and minimize transit times.

Key Research & Projects

CrashVision-DL: Traffic Severity Prediction System | *PyTorch, DeepInsight, EfficientNet*

- Engineered a Deep Learning pipeline transforming heterogeneous tabular data (US-Accidents, 7.7M records) into 120×120 feature images using t-SNE (DeepInsight).
- Designed a custom differentiable **Soft F-Beta loss** ($\beta = 2.0$) to prioritize recall for fatal crash detection, addressing severe class imbalance.
- Developed a hierarchical inference strategy decoupling binary classification from multi-class severity estimation.

Android Hardware Bridge over TCP/IP | *Linux, ADB, USB/IP, Shell*

- Architected a solution to convert rooted Android devices into headless USB servers using USB/IP protocols.
- Bypassed Android's restrictive sandboxing by injecting custom Linux binaries via ADB to expose hardware interfaces over Wi-Fi.

AI-Powered Scientific Content Pipeline | *Gemini Pro Vision, Edge-TTS, Asyncio*

- Built an ETL pipeline using PyMuPDF and Gemini Vision to convert scientific PDFs into bilingual audio notes.
- Developed a regex-based pre-processor to parse LaTeX formulas into natural language descriptions for TTS engines.
- Implemented an asynchronous architecture using **asyncio** for real-time audio generation and stateful checkpointing.

Youtube-Farsi-Dubber | *faster-whisper, yt-dlp, FFmpeg*

- Implemented an end-to-end automated dubbing pipeline: download video, transcribe (Whisper), translate, and synthesize audio (Edge-TTS) with auto-sync timing.
- Optimized processing speed using parallel execution for large batch operations.

EagleEye-Scraper | *Selenium, OpenCV, Tesseract*

- Developed a resilient CV-assisted web scraper using adaptive thresholding and visual-state monitoring to bypass anti-bot mechanisms.

Honors & Awards

i-Comp-Data Fellowship: Inter-faculty award promoting cross-disciplinary research projects in Computation and Data.

Top 1% Rank in National University Entrance Exam (among +150,000 candidates).

Merit-based Scholarship for outstanding academic performance at Amirkabir University.